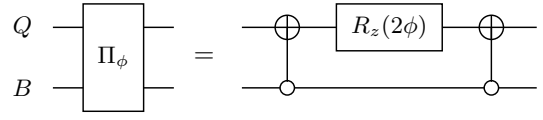
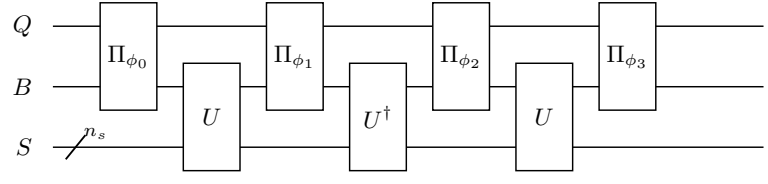
(a) Projector-controlled phase $\Pi_\phi = e^{i\phi(2\Pi - I)}$, $\Pi = |0\rangle\langle 0|_B$



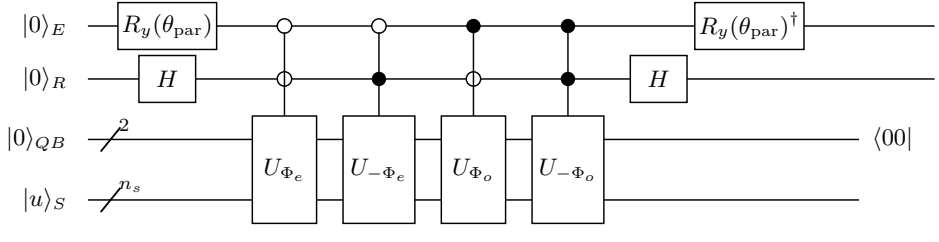
Q starts and ends in $|0\rangle$; phase $e^{+i\phi}$ when $B = 0$, $e^{-i\phi}$ when $B = 1$. For the one-qubit dilation this is just $e^{i\phi Z_B}$, which is what `apply_phase_sequence` applies.

(b) One phase sequence U_Φ of degree d ($d = 3$ drawn)



In every eigenplane of X, U acts as $R(x_j) = \begin{pmatrix} x_j & \sqrt{1-x_j^2} \\ \sqrt{1-x_j^2} & -x_j \end{pmatrix}$, so $\langle 0|_B U_\Phi |0\rangle_B = P_\Phi(X)$, $P_\Phi(x) = \left[e^{i\phi_0 Z} \prod_{k=1}^{d} R(x) e^{i\phi_k Z}\right]_{00}$ with $\mathrm{Re}\, P_\Phi = q_e$ or q_o (unit-peak even or odd part of $p/\max|p|$). Phases: `qsp_phases.find_phase_factors`, reflection convention. Here $U^\dagger = U$ (real symmetric).

(c) Real part and parity: the full phase-sequence block (paper 4.2; 4 ancillas E, R, Q, B)



R: H, sequence with $\pm\Phi$, H gives $\frac{1}{2}(P_\Phi + P_\Phi^*) = \mathrm{Re}\, P_\Phi$.

E: $R_y(\theta_{\mathrm{par}})|0\rangle = \sqrt{\frac{\alpha_e}{\alpha_e+\alpha_o}}|0\rangle + \sqrt{\frac{\alpha_o}{\alpha_e+\alpha_o}}|1\rangle$ selects even or odd.

Post-selected: $\frac{\alpha_e q_e(X) + \alpha_o q_o(X)}{\alpha_e + \alpha_o}|u\rangle = \frac{q(X)|u\rangle}{\alpha_e + \alpha_o}$

$P = \|q(X)u\|^2/(\alpha_e + \alpha_o)^2$; width $n_s + 4$; $4d$ calls of U as drawn

The package does not build this circuit: `QSVT.solve_phase_sequence` (centralise-2.1) runs (b) for the even and odd parts as dense linear algebra (`apply_phase_sequence`) and recombines $\alpha_e\, \mathrm{Re}\,\psi_e + \alpha_o\, \mathrm{Re}\,\psi_o$; (c) is the circuit that computes the same post-selected state. It was not run for the worked event (7.11).